

The problem: When you need guaranteed execution order

Introduction

What are workflow agents?

Workflow agents coordinate multiple sub-agents in a fixed, predictable pattern. Unlike the coordinator pattern from Part (where the LLM decides routing), workflow agents provide deterministic execution.

From ADK documentation

“Workflow agents are specialized agent types designed to orchestrate the execution of multiple sub-agents in specific patterns. They provide deterministic control flow.”

Three types:

- 1 SequentialAgent - Run agents one after another ($A \rightarrow B \rightarrow C$)
- 2 ParallelAgent - Run agents at the same time (A, B, C together)
- 3 LoopAgent - Repeat agents multiple times ($A \rightarrow B$, repeat

The problem:

When you need guaranteed execution order

Sometimes you need agents to run in a specific order, not based on LLM decisions:

```
Python
# What if you need guaranteed order?
# Step 1 MUST complete before Step 2
# Step 2 MUST complete before Step 3

# LLM-based routing might not guarantee this order
```

Use cases that need deterministic flow:

- Pipelines - Intake → process → respond (fixed order)
- Parallel tasks - Research from 3 sources at once (for speed)
- Refinement - Check quality, improve, repeat until good enough

